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$$\int_0^{\frac{1}{2}} \frac{\mathrm{Bgsin}\, x}{\sqrt{1-x^2}} dx = \int_0^{\frac{1}{2}} \mathrm{Bgsin}\, x d\left(\mathrm{Bgsin}\, x\right) \overset{t=\mathrm{Bgsin}\, x}{=} \int_0^{\frac{\pi}{6}} t dt = \left[\frac{t^2}{2}\right]_0^{\frac{\pi}{6}} = \frac{\frac{\pi^2}{36}}{2} = \frac{\pi^2}{72}$$

References